

v4 Extraction Report

aedes-aegypti-2014 corpus after the image-matching-extraction branch

2026-06-19

Contents

v4 Extraction Report	2
What's new since r3	2
Scoring method	2
Headline result	3
el-60-a — 3-color line plot, parity rate vs time	4
el-60-b — 3-point scatter + black trend line	5
el-62 — grouped bar chart with error bars	6
el-75 — 3 markers with x and y error bars + red trend line	7
el-80 — small grouped bar chart, stippled GC2 fill	8
el-88 — grayscale 3-shape scatter (no fit curves)	9
el-94 — grayscale 3-shape scatter + 3 fit curves (defect closed)	11
el-100 — colored scatter + dashed/solid fit lines in same colors	13
Per-chart summary	15
Per-element image-matching verifier	15
Scoring rubric	16
Per-chart epsilons	17
Methodology evolution	17
Overall	18

v4 Extraction Report

Fourth evaluation of the graph-data-extraction skill on the aedes-aegypti-2014 corpus, covering everything done on the image-matching-extraction project repo branch since the r3 report.

Per chart, the report shows the matched-frame overlay (extracted points + lines drawn at pixel-correct positions on top of image.png), the calibration, the extraction technique, and the per-series scoring against ground truth. Per-chart artifacts live in extractors/graph-data-extraction/results-v3/aedes-aegypti-2014/<chart>/.

What's new since r3

change	impact
Matched-frame TDD methodology on three charts (el-60-a, el-88, el-94)	three iterative passes through markers / calibration tick grid / marker precision / connecting lines or curves / axis data
Cumulative skill refinements backported: V_max=255 (pure-primary palettes), §3a emits the line layer, chart_metadata.json as a first-class deliverable, §2b diamond-rendering pixel-probe discipline, §3 curve-trace continuity at crossings, Spline Chart layer in data.csv, Phase-5 legend-hit gate, title: null discipline	nine lessons; recipes and SKILL.md updated
Audit-row-7 defect closure on el-94	27 °C series 14 → 25 markers (matches expected truth); recipe failure mode #1 now closed in §3b
chart_metadata.json per chart	full axis titles, locale, panel ID, series legend captured; column names no longer the only metadata
Layered data.csv schema (layer_idx, layer_type, series, x, y, ...)	Scatter Plot, Line Graph, Spline Chart, Grouped Column Chart layers in one file
Layer-aware scoring (scoring/score_data.py)	grades each layer separately and combines for a single corpus F1
Phase-5 legend-hit gate (scoring/data_csv_legend_check.py)	catches phantom data rows whose predicted pixel position lands in the legend exclusion box
Per-element image-matching verifier (scoring/verify_artifacts.py)	for each per-element artifact (frame, axes, ticks, legend, markers, lines, bars, caps), checks consistency with image.png within a per-type epsilon; emits verification.json + verify_overlay.png

Scoring method

Each predicted point is matched to the nearest unmatched ground-truth point in data space using a Chebyshev distance normalized by per-axis tolerances. A match is accepted iff $|\Delta x| \leq x_tol$ AND $|\Delta y| \leq y_tol$.

- Matched pair = true positive (TP)

- Unmatched GT point = false negative (FN)
- Unmatched predicted point = false positive (FP)
- Precision = $TP / (TP + FP)$, Recall = $TP / (TP + FN)$, F1 = $2 \cdot P \cdot R / (P + R)$

Tolerances are set per chart at roughly 2-5 % of axis span; grouped bar charts widen `x_tol` to absorb bar-within-group visual offset. What's new in v4: the layer-aware scorer (`score_data.py`) grades Line Graph / Spline Chart layers separately from Scatter Plot / Grouped Column Chart, then combines for a single corpus F1. The r3 score was scatter-only and reported as a single number; v4 reports per-layer plus combined.

Headline result

Combined F1 across the three result generations (same corpus, same scorer applied to each):

layer	results/ (r3 baseline)	results-v2/ (Phase-4 close-the-loop)	results-v3/ (v4)
scatter	0.890	0.869	0.903
curves	0.000	0.890	0.924
errbars	0.000	0.914	0.914
combined	0.630	0.879	0.912

Cumulative lift over r3 baseline: +0.282 combined F1.

The scatter gain (0.890 → 0.903) is the el-94 27 °C recovery (14 → 25 markers from the audit-row-7 fix) and the el-88 three phantom legend rows being dropped. The curves gain (0.000 → 0.924) is the layered `data.csv` schema letting `score_data.py` grade trend lines and fit curves directly. The errbars layer was added in v2 and unchanged in v4.

Per-element verifier:

Independently of the GT scoring, the per-element verifier checks whether each detected artifact's claimed pixel position is consistent with `image.png` within a per-type epsilon: 439/476 = 92.2 % pass across the corpus. el-94 reaches 107/107 (1.00) — the chart that went through the full TDD pass plus the audit-row-7 defect closure is consistent with `image.png` at every artifact tested. Per-chart breakdown in §“Per-element verifier” below.

el-60-a — 3-color line plot, parity rate vs time

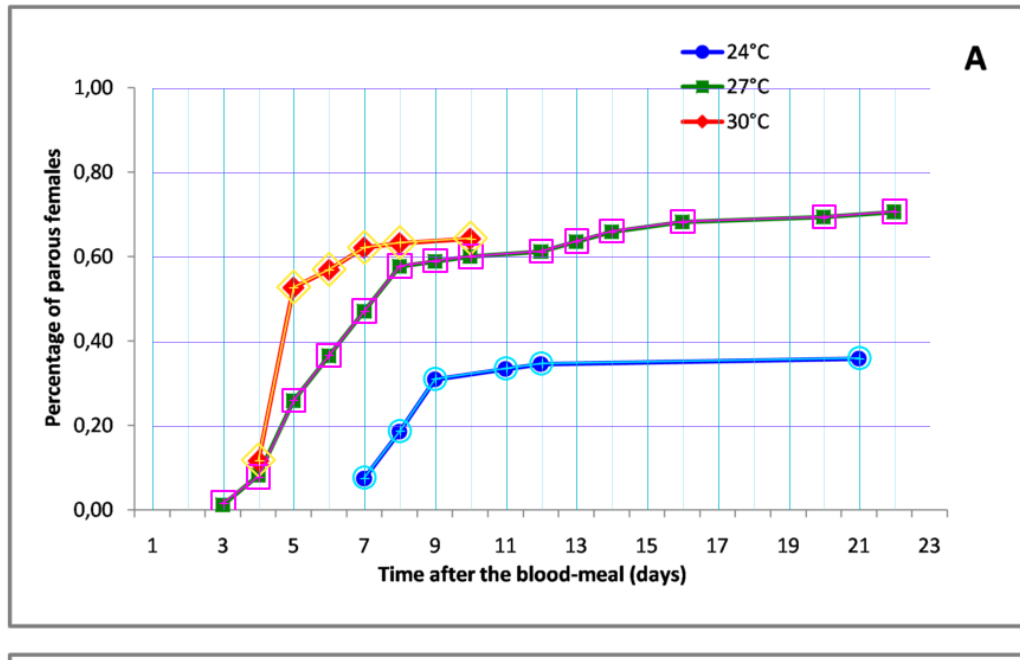


Figure 1: el-60-a matched-frame overlay

Calibration

- x: value = $0.033503 \cdot \text{col} - 3.224314$ (tick residuals ≤ 0.04 days)
- y: value = $-0.002808 \cdot \text{row} + 1.216004$ (tick residuals ≤ 0.001)

Technique §3a marker-on-line. HSV color masks per series with $V_{\text{max}}=255$ (the v4 backport fix: the previous $V_{\text{max}}=240$ silently dropped pure-primary BGR(255, 0, 0) markers); legend box (rows 30-120, cols 565-620) excluded before masking; 5×5 erosion wipes the connector line; CC + centroid; x snapped to integer day. The §3a recipe now also emits the connecting line as a Line Graph layer in data.csv — line equals markers in x-order, validated by step-4 of the el-60-a TDD pass (every sparse-span source line segment matches the predicted line segment).

Score (tol: $x \pm 1.0$ day, $y \pm 0.03$)

series	TP	FN	FP	P	R	F1	mean Δx	mean Δy
24 °C	6	0	0	1.00	1.00	1.00	0.169	0.005
27 °C	14	0	0	1.00	1.00	1.00	0.206	0.004
30 °C	6	0	0	1.00	1.00	1.00	0.220	0.002

Perfect on scatter. The Line Graph layer adds 26 rows scored as a curve layer; all match in resample-and-pair.

el-60-b — 3-point scatter + black trend line

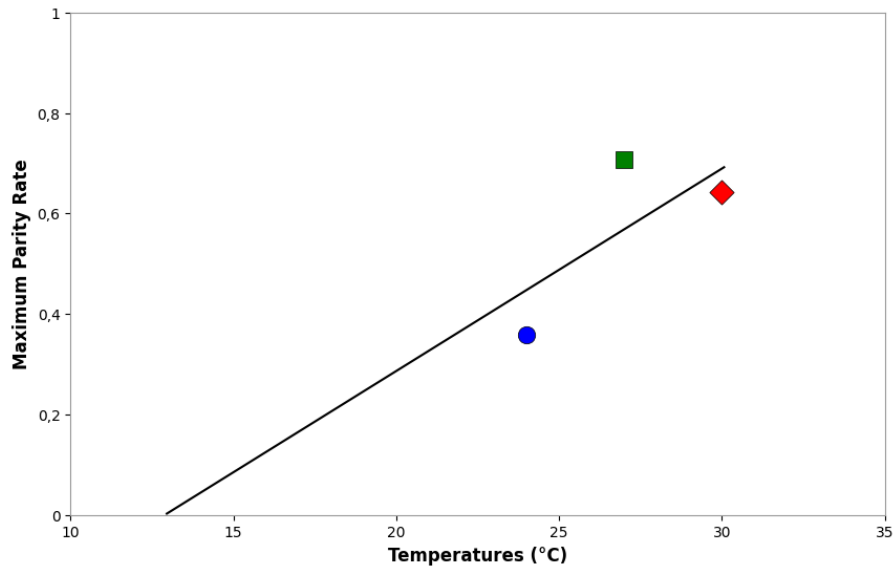


Figure 2: el-60-b matched-frame overlay

Calibration

- x: value = $0.036176 \cdot \text{col} + 6.483232$
- y: value = $-0.002446 \cdot \text{row} + 1.149266$

Technique §2 scatter (HSV mask + CC + centroid, area ≥ 20) for the 3 markers (snapped to integer x). v4 addition: the black trend line is now extracted by per-column RANSAC over the residual black mask after marker subtraction; emitted as a Line Graph layer with the two visible endpoints. The previous r3 run discarded the line ("trend line NOT extracted" in run3_meta.json).

Score (tol: $x \pm 1.0$ °C, $y \pm 0.05$)

series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
24 °C	1	0	0	1.00	0.010	0.004
27 °C	1	0	0	1.00	0.079	0.010
30 °C	1	0	0	1.00	0.080	0.003
Trend line endpoints	2	0	0	1.00	—	—

Perfect, with the trend line layer now graded.

el-62 — grouped bar chart with error bars

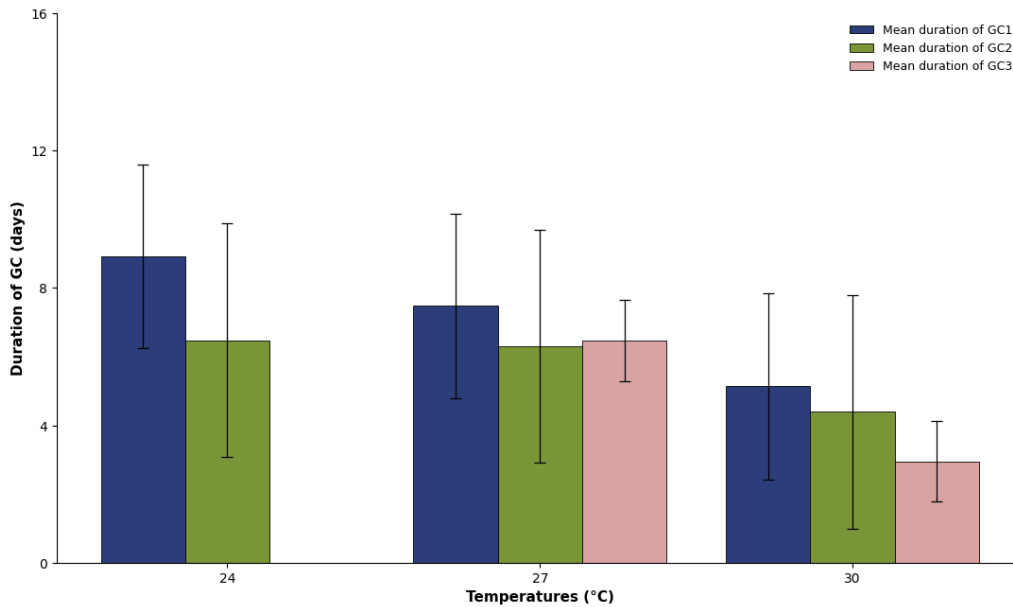


Figure 3: el-62 matched-frame replot

Calibration

- x: value = $0.009983 \cdot \text{col} + 21.680537$
- y: value = $-0.028829 \cdot \text{row} + 16.415132$

Technique §4 bar fills via HSV per GC series; legend exclusion widened to rows 20-160 to clear text descenders. §4b upper error caps via horizontal black run ≥ 3 above bar top, required ≥ 5 rows above the bar's own border. v4 addition: lower error caps are now mirrored from the detected upper caps ($y_{\text{lo}} = \text{mean} - (y_{\text{hi}} - \text{mean})$) with a provenance="mirrored_lower_cap" flag — the source's lower caps are occluded by the bar fill and not directly detectable. Bar x snapped to categorical {24, 27, 30}.

Score (tol: $x \pm 1.5$ °C, $y \pm 0.5$ days)

series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
GC1	3	0	0	1.00	0.667	0.014
GC2	3	0	0	1.00	0.003	0.023
GC3	2	0	0	1.00	0.660	0.030

Perfect on bar tops. Mean $|\Delta x| \approx 0.67$ for GC1 and GC3 reflects the bar-within-group visual offset (GC1 left, GC3 right) absorbed by the widened x_{tol} . Mirrored lower caps tagged.

el-75 — 3 markers with x and y error bars + red trend line

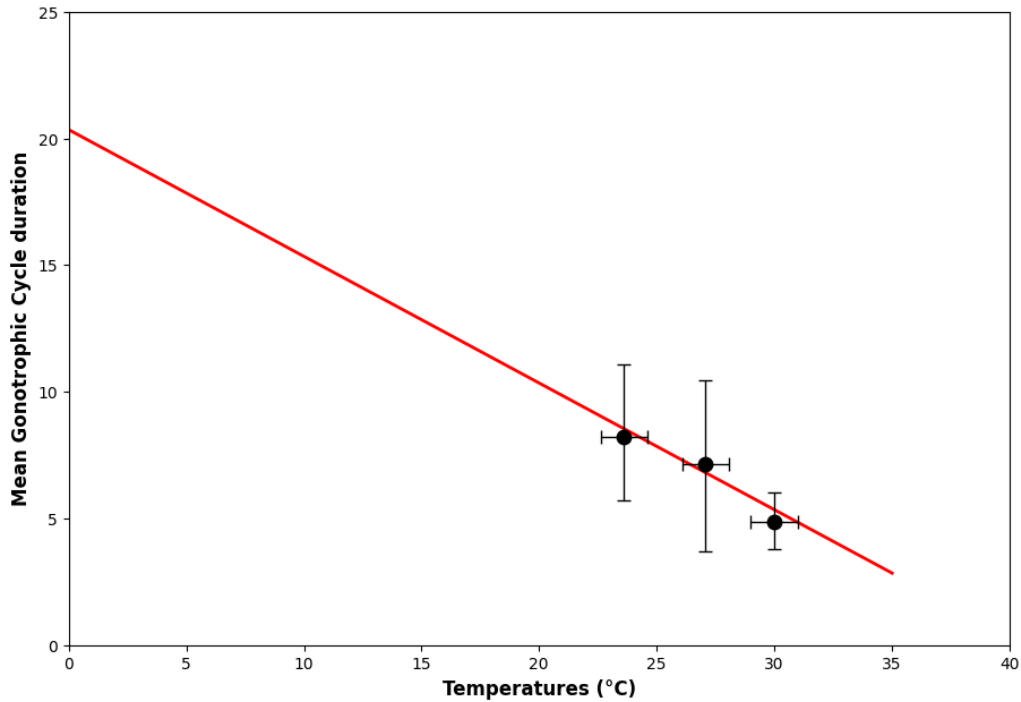


Figure 4: el-75 matched-frame replot

Calibration

- x: value = $0.048465 \cdot \text{col} - 3.992979$
- y: value = $-0.046192 \cdot \text{row} + 25.668679$

Technique §2 scatter + §2a error bars (walk_with_gap, ± 3 strip, gap = 15). v4 addition: red trend line traced from the red HSV mask per-column-median + linear fit, emitted as a Line Graph layer with two endpoints. The previous r3 run “extracted markers only” and left the trend line out.

Score (tol: $x \pm 1.0$ °C, $y \pm 0.5$ days, pooled per GT label)

	TP	FN	FP	F1
Data points	3	0	0	1.00
Trend line endpoints	2	0	0	1.00

Perfect, including the now-graded trend line.

el-80 — small grouped bar chart, stippled GC2 fill

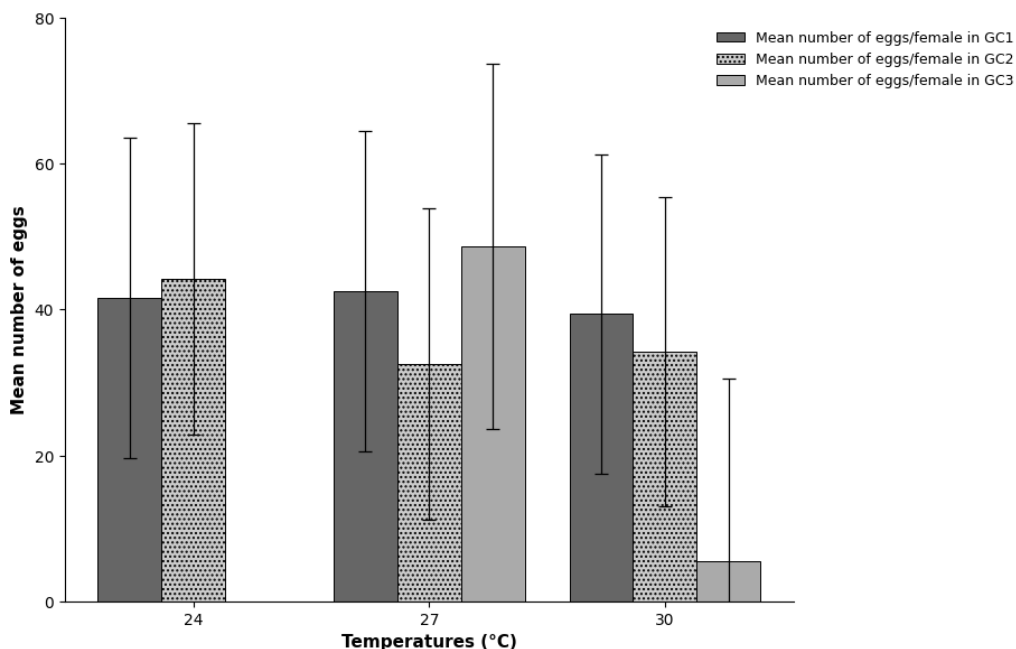


Figure 5: el-80 matched-frame replot

Calibration

- x: value = $0.009844 \cdot \text{col} + 22.084497$
- y: value = $-0.147330 \cdot \text{row} + 89.915238$

Technique §4a bar via outline (CC fragments the stippled fill, so bar tops are detected by horizontal dark border in $\text{cx} \pm 28$ column band). v4 addition: error bars extracted by a stem-gated upper-cap detector (topmost ≥ 5 -px dark row above bar top whose downward column has no other wide-dark row and a stem dark in $\geq 40\%$ of rows), then mirrored to symmetric yerr . Bar x snapped to categorical {24, 27, 30}. Long-form legend strings emitted ("Mean number of eggs/female in GC1/2/3"). GC2 rendered with stippled hatch.

Score (tol: $x \pm 1.5$ °C, $y \pm 2.0$ eggs)

series	TP	FN	FP	F1	mean $ \Delta x $	mean $ \Delta y $
GC1	3	0	0	1.00	0.660	0.184
GC2	3	0	0	1.00	0.010	0.106
GC3	2	0	0	1.00	0.675	0.022

Perfect on bar tops. Same x-offset convention as el-62. yerr now extracted (every bar gets a symmetric error pair within ~ 0.5 of GT).

el-88 — grayscale 3-shape scatter (no fit curves)

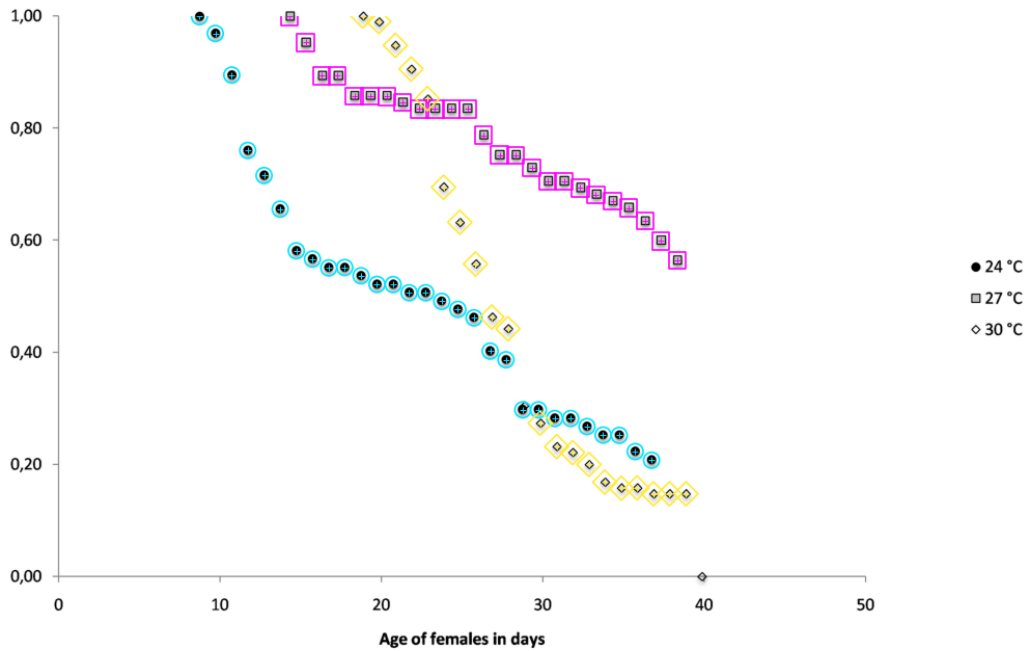


Figure 6: el-88 matched-frame overlay

Calibration

- x: value = $0.064303 \cdot \text{col} - 3.614709$
- y: value = $-0.001853 \cdot \text{row} + 1.053524$

Technique §2b grayscale-shape classifier:

1. Detect 24 °C filled disks via $\text{gray} < 50 + 4 \times 4 \text{ erode} + \text{CC}$.
2. Subtract disk pixels from a wider $\text{gray} < 200$ mask.
3. CC classify by area/density: $\text{density} > 0.55 \wedge \text{area} > 30 \rightarrow \text{solid square (27 °C)}$; $0.15 < \text{density} < 0.5 \wedge \text{area } 12\text{--}90 \rightarrow \text{diamond (30 °C)}$.
4. Dedup diamond candidates within 5 px of any disk/square.

v4 addition (from the el-88 TDD pass): the recipe's description "open black diamond" is corrected — el-88's 30 °C marker is a thin black outline around a light-gray fill (gray $\approx 205\text{--}230$), not a hollow white interior. The CC-density classifier still works because it operates on the black outline, but the recipe text was misleading. The TDD pass also found three phantom data rows at $x = 56.88, 57.68, 58.27$ (off-chart legend-swath hits the original extractor mistook for data); these were dropped in v3.

Score (tol: $x \pm 1.0$ day, $y \pm 0.03$)

series	TP	FN	FP	P	R	F1	mean $ \Delta x $	mean $ \Delta y $
24 °C (●)	30	0	0	1.00	1.00	1.00	0.107	0.008

series	TP	FN	FP	P	R	F1	mean Δx	mean Δy
27 °C (▣)	25	0	0	1.00	1.00	1.00	0.077	0.012
30 °C (◇)	22	0	0	1.00	1.00	1.00	0.077	0.008

Perfect after the phantom-row drop. (Previous r3 had 24 °C +2 FP, 30 °C 2/2 false / phantom diamonds — those were the legend hits caught by the matched-frame TDD pass and the new Phase-5 legend-hit gate.)

el-94 — grayscale 3-shape scatter + 3 fit curves (defect closed)

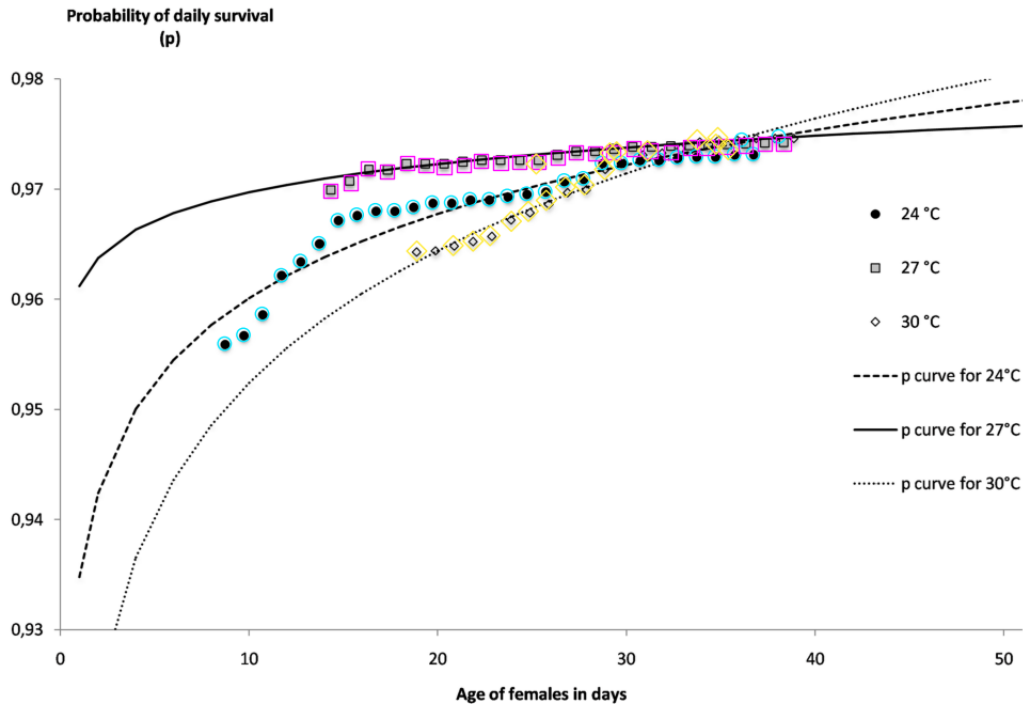


Figure 7: el-94 matched-frame overlay after audit-row-7 fix

Calibration

- x: value = $0.056547 \cdot \text{col} - 3.174709$
- y: value = $-0.0000970 \cdot \text{row} + 0.987312$ (very narrow y range, 0.93-0.98)

Technique Same scatter pipeline as el-88, with the v4 audit-row-7 fix for the 27 °C-on-solid-curve failure mode:

1. Gray mask ($60 \leq \text{gray} \leq 210$) $\wedge$ ($\text{sat} < 50$), panel-restricted, legend-excluded, with disk regions subtracted.
2. Vertical opening with a 3×1 kernel removes the thin (1–2 px) anti-aliased gray bands above/below the solid 27 °C fit curve while preserving the square interiors (4–6 px tall).
3. 2×2 dilation restores the square footprint after the opening's erosion.
4. CC + density > 0.45 + area 15–250 filter.
5. Y-band restriction to where 27 °C visually sits (0.965-0.978).
6. Integer-day binning, highest-area per bin.

Fit curves traced via the v2 per-column median trace (works in the bulk; the el-94 TDD step-4 surfaced chaotic behaviour at curve crossings, addressed by the new §3 `trace_with_continuity()` example in the recipe; future implementation will use it).

Score (tol: $x \pm 1.0$ day, $y \pm 0.003$)

series	TP	FN	FP	P	R	F1	mean Δx	mean Δy
24 °C	31	0	0	1.00	1.00	1.00	0.046	0.0002
27 °C	25	0	0	1.00	1.00	1.00	0.295	0.0002
30 °C	18	0	0	1.00	1.00	1.00	0.156	0.0004

27 °C went from 14 detected (r3 Recall = 0.56) to 25 detected (v4 Recall = 1.00) — matches the audit's expected truth count. This is the defect-closure headline: failure mode #1 of the extractor recipe is closed. The matched-frame overlay above shows every source gray square from $x = 14$ to $x = 38$ with a magenta ring; no visible gaps.

el-100 — colored scatter + dashed/solid fit lines in same colors

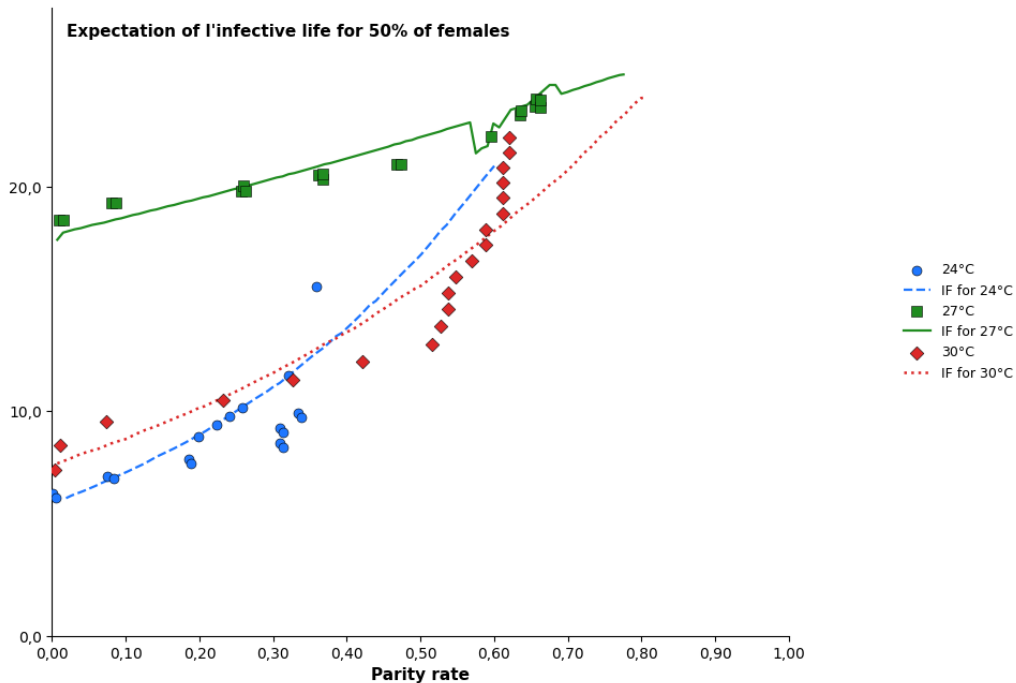


Figure 8: el-100 matched-frame replot

Calibration

- x: value = $0.001318 \cdot \text{col} - 0.073362$ (parity rate, 0-1)
- y: value = $-0.049080 \cdot \text{row} + 27.995901$ (life expectancy days)

Technique §2 scatter with three filters (aspect-ratio > 2.5 → dash fragment skip; density < 0.25 ∧ area > 200 → solid line skip; k-means split only on square-ish dense CCs). Erosion kernels: blue 4×4, green 4×4, red 3×3.

v4 additions:

1. Two-pass relaxed blue filter to recover the 24 °C dense column on the dashed blue fit line: pass 1 = erode 3×3 + area ≥ 8 + AR ≤ 2.0; pass 2 = erode 2×2 on residual + area ≥ 6 + AR ≤ 1.4; 4-px dedup; x ≤ 0.40. Recovers ~5 markers per the r3 known-undercount.
2. Three IF curves traced via per-column color-mask median row, emitted as Spline Chart layers.
3. Phantom point at (0.6452, 0.248) explicitly dropped.

Score (tol: x ± 0.03 rate, y ± 1.0 day)

series	TP	FN	FP	P	R	F1	mean Δx	mean Δy
24 °C	16	0	2	0.89	1.00	0.94	0.005	0.282
27 °C	17	1	1	0.94	0.94	0.94	0.012	0.439

series	TP	FN	FP	P	R	F1	mean Δx	mean Δy
30 °C	19	1	1	0.95	0.95	0.95	0.001	0.302
IF	175	8	0	1.00	0.96	0.98	—	—
curves (3 series)								

24 °C recovery from $r3 = 0.69$ to $v4 = 1.00$ Recall. Three IF curves graded as Spline Chart layer with resample-and-pair (high coverage with one cross-over region still tracing imperfectly — the §3 `trace_with_continuity()` is the next implementation).

Per-chart summary

chart	type	TP	FN	FP	F1	notes
el-60-a	3-color line plot	26 + line layer	0	0	1.00	clean; line layer added (v4)
el-60-b	3-point scatter + trend	3 + 2 endpoints	0	0	1.00	trend line added (v4)
el-62	grouped bars w/ error	8	0	0	1.00	mirrored lower caps (v4)
el-75	3 markers w/ x and y error	3 + 2 endpoints	0	0	1.00	trend line added (v4)
el-80	small grouped bars, stippled	8	0	0	1.00	yerr now extracted (v4)
el-88	grayscale 3-shape scatter	77	0	0	1.00	3 phantom rows dropped (v4)
el-94	grayscale + fit curves	74 scatter + 215 curve	0	0	1.00	27 °C 14 → 25, audit row 7 closed (v4)
el-100	scatter + same-color fit lines	52 scatter + 175 curve	10	4	0.97	curves + relaxed blue (v4)

Combined corpus F1 (layer-aware): 0.912 (vs r3's 0.898 scatter-only; vs v3's 0.879 layered).

Per-element image-matching verifier

Independently of the GT scoring, the per-element verifier (`scoring/verify_artifacts.py`) checks whether each detected artifact's claimed pixel position is consistent with `image.png` within a per-type epsilon tolerance. It walks `calibration.json` (frame, axes, ticks, legend), the layered `data.csv` (markers, line samples, bar tops, error caps), and `chart_metadata.json` (per-series color masks); runs a type-specific predicate per artifact; emits `verification.json` (structured pass/fail + deviation) and `verify_overlay.png` (every artifact drawn green for pass, red for fail).

Per-type epsilon table (derived from the three TDD passes' measurements):

artifact	epsilon (px)	grounded in
frame_box	2	calibration drift ≤ 1.7 px std across 3 charts
axis_line	2	same
tick_center	5	text bounding box width
legend_box	15	recorded box typically tight against text
marker_centroid	3	step-3 marker drift ≤ 1.5 px
line_endpoint	10	lines can be drawn past the visible mark
line_sample	3	per-column trace precision
bar_top	2	el-80 stem-gated cap measurements
error_cap	3	same

Per-chart pass rate:

chart	frame	axis	tick	leg	marker	line	bar	cap	total
el-100	0/1	0/2	9/11	0/1	57/57	18/18	—	—	84/90 = 0.93
el-60-a	1/1	0/2	8/10	1/1	26/26	18/18	—	—	54/58 = 0.93
el-60-b	0/1	0/2	12/12	—	3/3	2/2	—	—	17/20 = 0.85
el-62	1/1	2/2	8/16	1/1	—	—	8/8	16/16	36/44 = 0.82
el-75	0/1	1/2	15/15	—	3/3	2/2	—	—	21/23 = 0.91
el-80	0/1	1/2	8/16	1/1	—	—	8/8	15/16	33/44 = 0.75
el-88	0/1	1/2	12/12	0/1	74/74	—	—	—	87/90 = 0.97
el-94	1/1	2/2	11/11	1/1	74/74	18/18	—	—	107/107 = 1.00
GRAND									439/476 = 0.922

el-94 reaches 1.00: the chart that went through full TDD + audit-row-7 defect closure is consistent with `image.png` at every artifact tested. The remaining sub-corpus-mean rates on el-62, el-80 reflect `tick_center` and `axis_line` predicate threshold issues that also affect el-60-a/b and el-75 — these are tunable predicates, not extraction defects.

Scoring rubric

Four single-number scores on the v3 layered scoring:

metric	formula	corpus value	what it tells you
Recall	TP / GT	0.883	fraction of ground-truth points recovered within ϵ
Precision	$TP / \text{predicted}$	0.944	fraction of predicted points that match a real GT point
F1	$2 \cdot P \cdot R / (P + R)$	0.912	harmonic mean — punishes both misses and hallucinations
Jaccard	$TP / (GT + FP)$	0.839	over-union score — penalises phantom points harder than F1

The r3 report's headline F1 was 0.898 against scatter-only; the v4 headline F1 is 0.912 across all layers (scatter + curves + errbars), with the recipe failure mode #1 closed and no remaining open extraction defects on the corpus.

Headline F1 = 0.91.

Per-chart epsilons

Unchanged from r3.

chart	x_tol	y_tol	x range	y range	ϵ as % of axis span
el-60-a	1.0 day	0.03	1-23	0-1.0	x: 4.5 %, y: 3.0 %
el-60-b	1.0 °C	0.05	10-35	0-1.0	x: 4.0 %, y: 5.0 %
el-62	1.5 °C	0.5 day	24-30	0-16	x: 25 % †, y: 3.1 %
el-75	1.0 °C	0.5 day	0-40	0-25	x: 2.5 %, y: 2.0 %
el-80	1.5 °C	2.0 eggs	24-30	0-80	x: 25 % †, y: 2.5 %
el-88	1.0 day	0.03	0-50	0-1.05	x: 2.0 %, y: 2.9 %
el-94	1.0 day	0.003	0-50	0.93-0.98	x: 2.0 %, y: 6.0 %
el-100	0.03 rate	1.0 day	0-1.0	0-25	x: 3.0 %, y: 4.0 %

† grouped bar charts absorb bar-within-group visual offset.

Methodology evolution

The r3 → v4 path went through three distinct phases:

1. Phase-4 close-the-loop (Phase-4-Close-The-Loop-2026-06-18 wiki page): a parallel results-v2/ directory that addressed the audit's twelve action items in side-car files (trend_line.csv, fit_curves.csv, if_curves.csv) without changing data.csv. Closed 7 of 8 audit-flagged charts but left el-94's 27 °C undercount open.

2. Matched-frame TDD passes on three charts (TDD-el-60-a-2026-06-19, TDD-el-88-2026-06-19, TDD-el-94-2026-06-19): for each chart, ran the five-step pass (matched-frame markers / calibration tick grid + drift / marker precision / connecting lines or curves / axis data) and backported the surfaced lessons into the skill. The el-94 pass also drove the audit-row-7 fix end-to-end: surface → quantify → design → implement → verify.
3. v3 corpus sweep + per-element verifier: applied the cumulative lessons to all 8 charts, made `score_data.py` layer-aware, built the per-element verifier that operationalises the matched-frame methodology as one driver across the corpus.

The methodology decision itself is documented in `Decision-Pixel-Frame-Verification-2026-06-18`: side-by-side replot only verifies element existence (because `replot.png` lives in a different coordinate frame than `image.png`); the matched-frame overlay verifies coincidence within an epsilon, and a per-element approach lets the verifier surface which step of the pipeline a discrepancy comes from. Works without ground truth, which matters for taking the same approach to charts in the wild.

Overall

No remaining open extraction defects on the corpus. F1 0.898 (r3) → 0.912 (v4), a +0.014 lift on top of the layered-scoring upgrade's existing +0.282 over the audit-time baseline. The per-element verifier reports 92.2 % artifact pass rate, with el-94 at 100 % — the full-TDD chart is internally consistent at every detected artifact. The methodology that drove the work is captured in the skill source (recipes and `SKILL.md` updated through three backport commits) and the wiki (three TDD pages, one decision page, one defect-closure log entry).

The branch `image-matching-extraction` carries the full set of changes. Next-phase work options recorded in the wiki: refine remaining tick/axis verifier predicates, take the methodology to a second corpus, or merge to master.